

BCAC 0814: SOFTWARE LAB USING PYTHON

Objective: The course is designed to provide basic knowledge of Python. Python programming is intended for software engineers, system analysts, program managers and user support personnel who wish to learn the Python programming language.

Credits: 02

L-T-P: 0-0-4

Module No.	Content	Lab Hours
I	1. Write a menu driven program to convert the given temperature from Fahrenheit to Celsius and vice versa depending upon user's choice. 2. WAP to calculate total marks, percentage and grade of a student. Marks obtained in each of the three subjects are to be input by the user. Assign grades according to the following criteria: - Grade A: Percentage ≥ 80 - Grade B: Percentage ≥ 70 and < 80 - Grade C: Percentage ≥ 60 and < 70 - Grade D: Percentage ≥ 40 and < 60 - Grade E: Percentage < 40 3. Write a menu-driven program, using user-defined functions to find the area of rectangle, square, circle and triangle by accepting suitable input parameters from user. 4. WAP to display the first n terms of Fibonacci series. 5. WAP to find factorial of the given number. 6. WAP to find sum of the following series for n terms: $1 - 2/2! + 3/3! - \dots - n/n!$. 7. WAP to calculate the sum and product of two compatible matrices.	20
II	8. Write a menu-driven program to create mathematical 3D objects I. curve II. sphere III. cone IV. arrow V. ring VI. Cylinder. 9. WAP to read n integers and display them as a histogram. 10. WAP to display sine, cosine, polynomial and exponential curves. 11. WAP to plot a graph of people with pulse rate p vs. height h. The values of p and h are to be entered by the user. 12. WAP to calculate the mass m in a chemical reaction. The mass m (in gms) disintegrates according to the formula $m = 60/(t+2)$, where t is the time in hours. Sketch a graph for t vs. m, where $t \geq 0$. 13. A population of 1000 bacteria is introduced into a nutrient medium. The population p grows as follows: $P(t) = (15000(1+t))/(15 + e)$ where the time t is measured in hours. WAP to determine the size of the population at given time t and plot a graph for P vs t for the specified time interval. 14. Input initial velocity and acceleration, and plot the following graphs depicting equations of motion: I. velocity wrt time ($v = u + at$) II. distance wrt time ($s = ut + 0.5at^2$) III. distance wrt velocity ($s = (v^2 - u^2)/2a$)	20

Reference Books:

- Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 2nd edition, Updated for Python 3, Shroff/O'Reilly Publishers, 2016. (<http://greenteapress.com/wp/thinkpython/>)
- Guido van Rossum and Fred L. Drake Jr, "An Introduction to Python – Revised and updated for Python 3.2, Network Theory Ltd., 2011.

- Charles Dierbach, "Introduction to Computer Science using Python: A Computational Problem-Solving Focus, Wiley India Edition, 2013.
- John V Guttag, "Introduction to Computation and Programming Using Python", Revised and expanded Edition, MIT Press, 2013
- Kenneth A. Lambert, "Fundamentals of Python: First Programs", CENGAGE Learning, 2012.

Outcome: After studying the subject, the students will be able to:

- CO1: To learn and understand Python programming basics.
- CO2: To learn and understand python looping, control statements and string manipulations.
- CO3: Students should be made familiar with the concepts of GUI controls and designing GUI applications.
- CO4: To learn and know the concepts of file handling, exception handling and database connectivity.